



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE489

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Project Title: Retail Shop Management & POS System

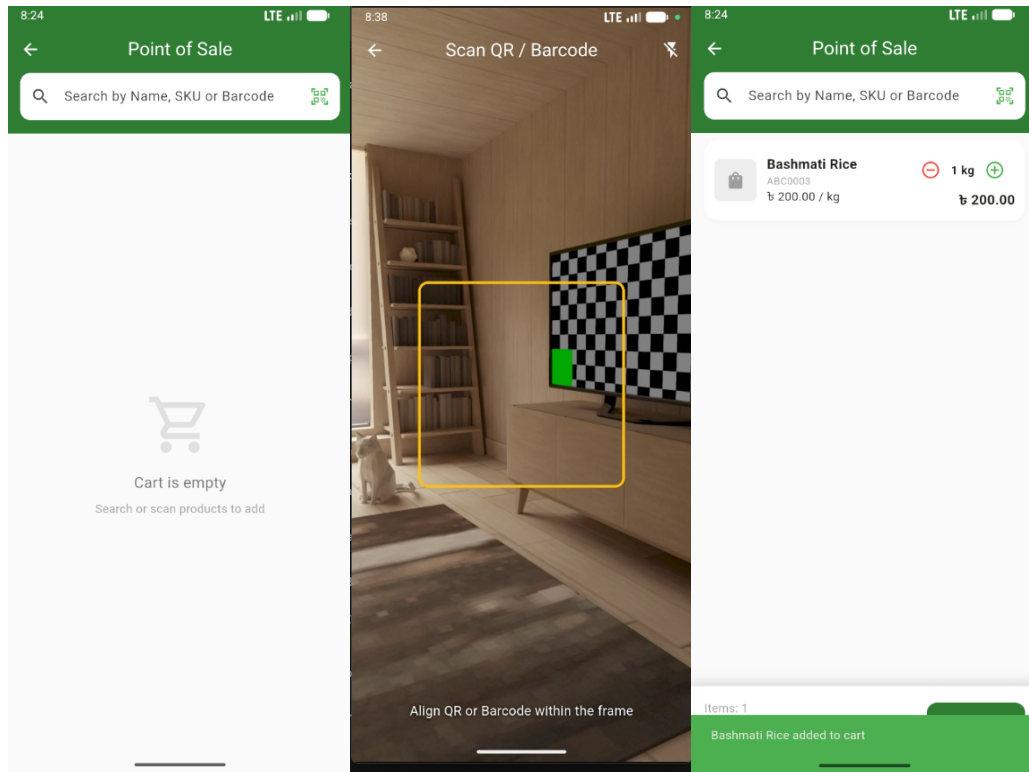
Project Features:

By the end of this course, implemented features:

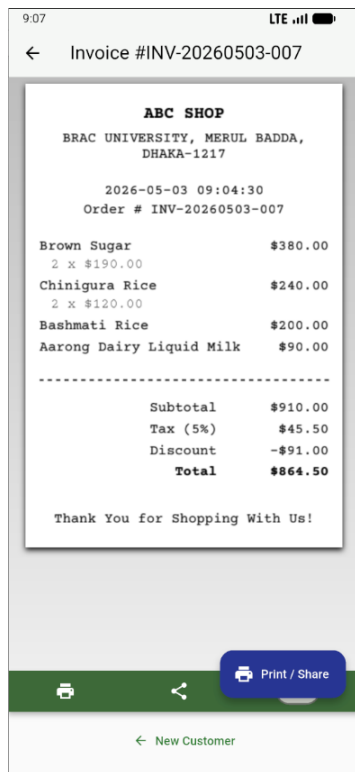
- a) **Role-Based Access Control:** Secure multi-tier login system (Admin, Owner, Manager, Staff) with specific dashboard access.
- b) **Product Management:** Complete module to add, edit, and manage products, pricing, and stock counts.
- c) **Real-time Inventory Tracking:** Automatic deduction of stock levels during transactions and automated stock restoration for voided orders.
- d) **Low-Stock Alerts:** Automated dashboard warnings and interactive adjustment sheets when items fall below a specific quantity to prevent stockouts.
- e) **Barcode Scanning:** Integration with the device camera to scan retail barcodes, rapidly identifying products and adding them to the bill.
- f) **Digital Billing & Checkout:** Dynamic cart system calculating totals, taxes, and discounts (coupons), seamlessly generating a shareable/printable PDF thermal invoice.
- g) **Sales Analytics & Reporting:** Real-time dashboards visualizing today's snapshot, top-selling products, and revenue series.
- h) **Customer Database Management:** Tracking walk-in vs. registered customers, their transaction history, and total spent.

Screenshots:

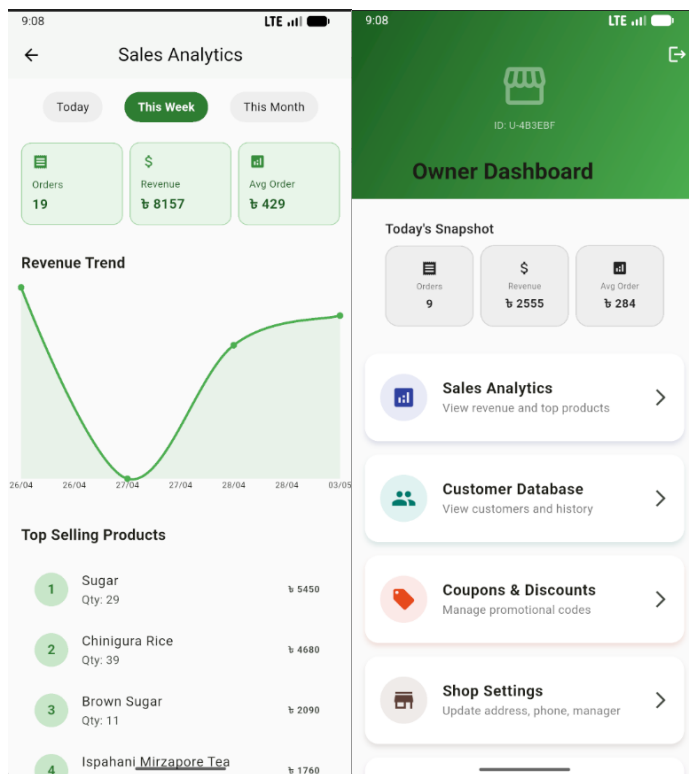
1. The POS Barcode Scanner / Cart Screen



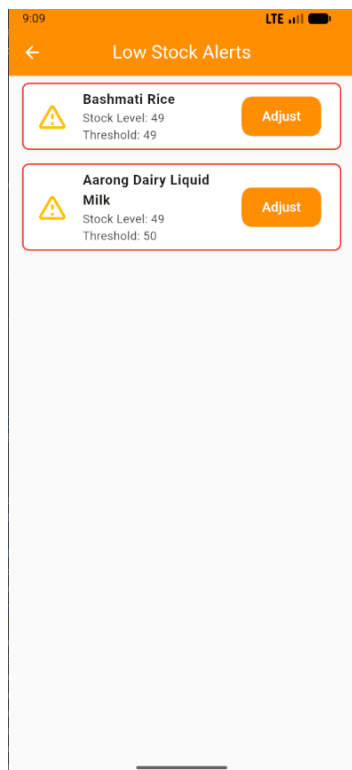
2. The PDF Invoice / Receipt Preview



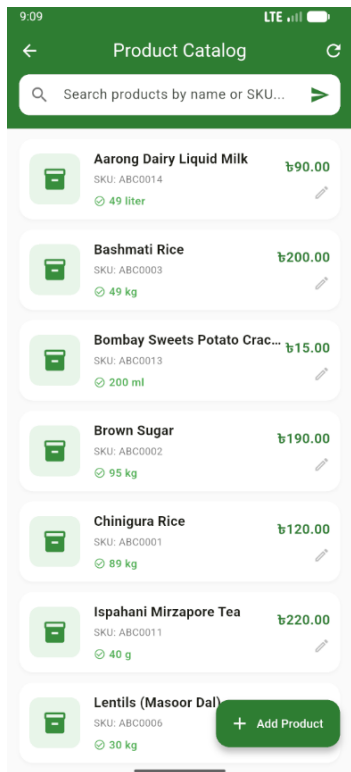
3. The Owner/Manager Analytics Dashboard



4. The Low-Stock Alerts Screen



5. The Product Management List



Online Resources used:

a) Reference:

- flutter.dev and pub.dev (for package documentation like **mobile_scanner**, **pdf**, **provider**)
 - <https://pub.dev/packages/pdf>
 - <https://pub.dev/packages/printing>
- expressjs.com and mongoosejs.com (for backend API development)
- Youtube videos:
 - <https://youtu.be/CzRQ9mnmh44?si=haymMeWaMNkMELGM>
 - <https://www.youtube.com/watch?v=uSljGJGS16w>
 - <https://www.youtube.com/watch?v=622vOT8A1J0>

b) *Stackoverflow or github links.*

- Stackoverflow for debugging Flutter layout issues and MongoDB aggregation queries.

- GitHub documentation:
 - https://github.com/juliansteenbakker/mobile_scanner
 - https://github.com/DavBfr/dart_pdf

Future Enhancements:

Following enhancement can be added to the current system which will improve the system.

- a) **Multi-Branch Synchronization:** Allowing super-admins to manage centralized inventory pushed out to multiple localized retail branches.
- b) **Offline Mode:** Enabling the POS system to function without an internet connection and seamlessly syncing transactions to the backend once the connection is restored.
- c) **AI Sales Forecasting:** Using historical transaction data to predict future sales trends and automatically suggest restock quantities.
- d) **Customer Loyalty Program:** A point-based reward system integrated into the checkout process based on a customer's total spent history.
- e) **Supplier & Purchase Orders:** A dedicated module to track suppliers, generate purchase orders, and automatically log incoming deliveries into the inventory.